

INST327 - Database Design and Modeling

Section 0203

Diabetes Patient Analysis Dataset

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Group G3

Thomas Tran, Alexander Lara, Ehsan Semati, Johanna Pentapaty

Project Proposal

Introduction:

Our dataset includes the demographics, health background, and medical history of individuals with different variations of diabetes which was sourced from multiple places such as clinical studies, public health databases, and hospital records which was last updated in October 2024. Specifically, there is a substantial and detailed record of different forms of diabetes, medical/clinical test results, lifestyle factors, and genetic profiles of each target in the dataset. With this data, specialized and diverse research can be executed on the various kinds of diabetes which will be further discussed later in this report. The dataset generally includes rows of individuals (targets) and their attributes that relate to their diabetes diagnosis.

This includes their actual diagnosis which can range from type 1 diabetes, Wolfram-Syndrome, LADA, gestational diabetes, type 3c diabetes, Neonatal diabetes mellitus, steroid-induced diabetes, etc. Related attributes to lifestyle factors of the target that attains to diabetes include physical activity, dietary healthiness, smoking and drinking activity, environmental/socioeconomic factors, and more. Each target's genetic profile is also in the dataset containing genetic markers, autoantibodies, family history with diabetes, genetic testing, etc. With these themes of data within the dataset, medical/clinical test results are the most important for comprehensive diabetes research. Some attributes within this theme include insulin levels, BMI, blood pressure, pulmonary function, digestive enzyme levels, urine test results, etc. which are all significant tests for individuals diagnosed with diabetes.

Our team chose this topic because of some of our family's histories on diabetes. Diabetes is so prevalent in most of our lives, and because of our genetic profile, we are also most likely to develop diabetes as well. This dataset is also a great reference for potential future professional projects for medical/clinical research company employers.

Target Audience:

Target groups that would benefit from using this dataset includes medical researchers, genetic researchers, people diagnosed with diabetes searching for a deeper analysis, public health professionals, pre-med students, and pregnancy researchers with each their own purpose. Medical researchers are able to use this dataset as a source to do statistical analysis, basing symptoms, diagnoses, and causes of different kinds of diabetes. Genetic researchers can analyze how important genetic factors are to developing a kind of diabetes. Public health professionals can study the data to strengthen decision-making in diabetes diagnoses. This reason would align with pre-med students as well for preparation to make diagnosis decisions. Pregnancy researchers can also use this due to the database's attributes of pregnancy history to examine who's at risk for gestational diabetes. Considering these possible stakeholders, these are our potential entities for the database to correlate.

Entities/Tables:

Targets	Genetics	Diagnosis	Lifestyle Factors
target_id	target_id	target_id	target_id
age	genetic_markers	diagnosis	environmental_factors
birth_weight			physical_activities
early_symptoms	family_history		dietary_habits
	genetic_testing		Socioeconomic factors
	ethnicity		smoking_status
			alcohol_consumption

General Test Results	Medical Test Results	Medical History	Pregnancy History
target_id	target_id	target_id	target_id
blood_pressure	insulin_levels	steroid_use_history	pregnancy_history
bmi	glucose_tolerance	cystic_fibrosis_diagnosis	pcos_history
waist_circumference	pancreatic_health		pregnancy_weight_gain

cholesterol_levels	pulmonary_function		prev_gestational_diabetes
blood_glucose_levels	neurological_assessments		
	liver_function_tests		
	digestive_enzyme_levels		
	Urine_test autoantibodies		

Non-included Entities:

With these entities, there will be subjects that won't be included into this database in respect of relevancy and viability. One entity that won't be included within the dataset would be glucose tests. This column within this table is already within both the general tests and medical tests table and won't be viable due to other tests metrics and tests also being significant to research in diabetes. Another table that won't be included is a Socio-Environmental table. This table would be redundant and its attributes would also attain the lifestyle factors table. Although it would be useful in research of societal factors leading to diabetes, the rest of the dataset wouldn't be relevant. Finally, another table that will not be included is the organs_health table which would include pulmonary, pancreatic, and liver functions health attributes. These would fall into the medical tests results table, and would also be redundant. A large medical test results table is capable of showing all specialized tests in order to collect and analyze medical data on targets in a more aggregated and efficient way.

Sample data:

Our dataset has a total of 70,000 rows of data in our dataset, meaning that there are 70,000 randomly selected patients in our dataset. I think a potential way of reducing the dataset is selecting a random fraction of the dataset, such as the first 10,000 rows, since the patients were randomly selected. Furthermore, we could choose to filter out a certain subset of data. We could filter by diagnoses type, leaving out very uncommon types of diagnoses. Or we could leave out certain age ranges, such as focusing on only young adults. We could also reduce the overall complexity of the dataset by removing certain columns. For instance, the column which tracks the diagnosis of cystic fibrosis (Cystic fibrosis diagnosis) may be redundant, because cystic

fibrosis is already displayed with the name of the diagnosis in the column which shows the diagnosis name(Target).

10 Potential Questions the Database can Answer

- What is the average age of patients diagnosed with each type of diabetes?
- Which type of diabetes has the highest average glucose levels?
- Which patients have both “high” BMI and glucose levels
- How many patients with each type of diabetes?
- What's the average age of each category of diagnosed diabetes?
- What's the average cholesterol of each type of diabetes?
- What socioeconomic status are a majority of the patients of each diabetes type?
- Do certain harmful lifestyle choices correlate with a certain type of diabetes?
- Do patients with a family history of diabetes have complications with birth and pregnancy?
- What diabetes is the most common in the dataset?

Diversity, Equity, and Inclusion Considerations:

The diabetes dataset included seems to be an inclusive information resource. This is specifically shown through including demographic variables such as age, ethnicity, lifestyle factors, and socioeconomic factors. The age range captures information from zero years all the way up to 80 and higher. These factors ensure that differences in the data are still able to be analyzed through various groups of people. Along with this, the addition of information related to pregnancies allows gestational diabetes to be studied, as it is a form first diagnosed during pregnancy (Mayo Clinic Staff 2025). As some major demographic information such as sex, access to healthcare, and disability statuses are not included, there may be some limits to the database in regards to its ability to completely represent both the social and demographic diversity of all diabetic patients.

There is an emphasis on both clinical and genetic data which may play a role in more biological reasoning for diabetes rather than social aspects. Even though there are a few columns in relation to lifestyle choices, there is still a lack of social determinants of one's health. In addition to this, the columns referring to pregnancy indicate that all of the subjects of this dataset

are females, leaving out a large portion of people with diabetes. Without proper socioeconomic variables, the dataset cannot factor in inequalities that play a role in diabetic risk.

Data Privacy, Fair Use, Other Ethical Considerations:

The information in this dataset is very sensitive, including genetic markers, pregnancy history, and results of tests. Even without the names, the information can still relate to actual people, especially genetic markers, which cannot be hidden. Therefore, the actual system must have security, restricted access, and rules for viewing the information. It also helps to have only the information that is actually needed. For instance, special genetic markers or special information about the pregnancy must only be included if they actually help the research and the individual agrees to have the information included.

We must also consider the sources for the sample information. If actual information from restricted sources such as patient records is used, it may violate copyright law. If actual information is used, the sources must be ones that allow the information to be used for research including the rules for sharing the information and the length of time the information can be kept. Even though the information is available to the general public, it is also important to verify that the information is reusable. This is to ensure that the research is conducted while maintaining the privacy of people and keeping the research ethical.

Citations:

Batra, A. (2024, September 1). Diabetes dataset. Kaggle.

<https://www.kaggle.com/datasets/ankitbatra1210/diabetes-dataset>

“Gestational Diabetes - Symptoms and Causes - Mayo Clinic.” *Mayo Clinic*, 30 May 2025,

www.mayoclinic.org/diseases-conditions/gestational-diabetes/symptoms-causes/syc-20355339.